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CCNA - Network Troubleshooting

- **Using Ping and TracerouteTheory :How Ping and Traceroute work.Practical :Use Ping and Traceroute to test connectivity.Task: Report findings for a simulated connectivity issue.**

1. **Open Command Prompt or Terminal and use the ping command to test connectivity to www.spotify.com. Note the number of packets sent, received, and the average response time.**

ANS.

- **Objective**
  To test network connectivity between the computer and www.spotify.com using the ping command.

- **Command Used — Windows**
  ping www.spotify.com

- **Sample Output**
  Pinging www.spotify.com [104.199.65.124] with 32 bytes of data:

  Reply from 104.199.65.124: bytes=32 time=38ms TTL=117
  Reply from 104.199.65.124: bytes=32 time=40ms TTL=117
  Reply from 104.199.65.124: bytes=32 time=39ms TTL=117
  Reply from 104.199.65.124: bytes=32 time=41ms TTL=117

  Ping statistics for 104.199.65.124:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
  Approximate round trip times in milli-seconds:
  Minimum = 38ms, Maximum = 41ms, Average = 39ms

Parameter	Result
Packets Sent	4
Packets Received	4
Packets Lost	0
Packet Loss	0%
Minimum Time	38 ms
Maximum Time	41 ms
Average Response Time	39 ms

2. Use the traceroute (tracert on Windows, traceroute on Linux/Mac) command to trace the path from your computer to www.flipkart.com. List the number of hops and identify the IP address of the first and last hop.

ANS.

- **Objective**
  To identify the path taken by network packets from the local computer to www.flipkart.com using the tracert command.

- **Command Used — Windows**
  tracert www.flipkart.com

- **Sample Output**
  Tracing route to www.flipkart.com [104.18.24.92]
  over a maximum of 30 hops:

  1 1 ms 1 ms 1 ms 192.168.1.1
  2 4 ms 5 ms 4 ms 10.10.0.1
  3 8 ms 7 ms 9 ms 100.64.0.1
  4 15 ms 14 ms 16 ms xxx.xxx.xxx.xxx
  5 24 ms 23 ms 25 ms xxx.xxx.xxx.xxx
  6 31 ms 30 ms 32 ms xxx.xxx.xxx.xxx
  7 38 ms 37 ms 39 ms 104.18.24.92

  Trace complete.

Parameter	Result
Destination	www.flipkart.com
Number of Hops	7
First Hop	192.168.1.1
Last Hop	104.18.24.92
Command	tracert www.flipkart.com

- **Explanation**
  The first hop, 192.168.1.1, represents the local network gateway/router. The following hops represent intermediate routers through which the packets travel. The final hop represents the destination reached by the trace.

3. Simulate a connectivity issue by pinging a non-existent website (for example, ping www.fakezomato12345.com) and write down the error message you receive. Hint: Compare the output to a successful ping and describe the difference in your findings.

ANS.

- **Objective**
  To test connectivity to a non-existent domain and compare the result with a successful ping.

- **Command Used**
  ping www.fakezomato12345.com

- **Sample Output**
  Ping request could not find host www.fakezomato12345.com.
  Please check the name and try again.

Feature	Successful Ping	Non-existent Website
Domain	www.spotify.com	www.fakezomato12345.com
DNS Resolution	Successful	Failed
Server IP obtained	Yes	No
Reply received	Yes	No
Packet statistics	Displayed	Not reached
Main issue	No connectivity problem	Domain cannot be resolved

- **Explanation**
  In the successful Spotify test, the domain name was resolved to an IP address and the computer received ICMP replies.

  For the fake website, the domain does not exist or cannot be resolved by the configured DNS server. Therefore, Windows cannot determine the destination IP address and displays:

  Ping request could not find host

4. Imagine you are unable to access www.youtube.com from your system. Use both ping

and traceroute commands to diagnose the issue and write a short report describing where the connection fails and what the possible reasons could be.

ANS.

- **Objective**
  Assume that www.youtube.com cannot be accessed. The ping and tracert commands can be used together to determine whether the problem is related to DNS resolution, local connectivity, routing, or the destination network.

- **Step 1: Test Using Ping**

  - **Command**
    ping www.youtube.com

  - **Possible Result A — Ping Successful**
    Pinging www.youtube.com [142.250.xxx.xxx] with 32 bytes of data:

    Reply from 142.250.xxx.xxx: bytes=32 time=35ms TTL=117
    Reply from 142.250.xxx.xxx: bytes=32 time=37ms TTL=117
    Reply from 142.250.xxx.xxx: bytes=32 time=36ms TTL=117
    Reply from 142.250.xxx.xxx: bytes=32 time=38ms TTL=117

    Ping statistics:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

- **Step 2: Trace the Route**

  - **Command**
    tracert www.youtube.com

  - **Example Result**
    Tracing route to www.youtube.com [142.250.xxx.xxx]

    1 1 ms 1 ms 1 ms 192.168.1.1
    2 5 ms 4 ms 5 ms 10.10.0.1
    3 10 ms 9 ms 11 ms xxx.xxx.xxx.xxx
    4 18 ms 17 ms 19 ms xxx.xxx.xxx.xxx
    5 25 ms 24 ms 26 ms xxx.xxx.xxx.xxx
    6 34 ms 33 ms 35 ms xxx.xxx.xxx.xxx
    7 40 ms 39 ms 41 ms 142.250.xxx.xxx

    Trace complete.

Test	Observation	Possible Meaning

ping www.youtube.com	Replies received	Basic connectivity is working
ping www.youtube.com	Request timed out	ICMP may be blocked or packets may be lost
ping www.youtube.com	could not find host	DNS resolution problem
tracert stops at first hop	Possible local router/network problem	
tracert stops after ISP hops	Possible routing/ISP issue	
tracert reaches destination	Route exists; problem may be application-level	
Browser fails but ping works	Possible DNS, HTTPS, browser, proxy, or filtering issue	

- **Possible Reasons for the Problem**

1. **DNS problem** — The system cannot resolve www.youtube.com correctly.
2. **Local network problem** — The computer may not have a valid IP address, gateway, or network connection.
3. **Router problem** — The default gateway may be unavailable.
4. **ISP/routing problem** — Packets may fail somewhere between the local network and the destination.
5. **Firewall/security filtering** — ICMP or web traffic may be blocked.
6. **Proxy/VPN configuration** — Incorrect proxy or VPN settings may prevent browser access.
7. **Application-level problem** — Ping may work even when HTTPS/web access has another problem.
8. **Destination-side ICMP filtering** — A website can be reachable while ignoring ping requests.